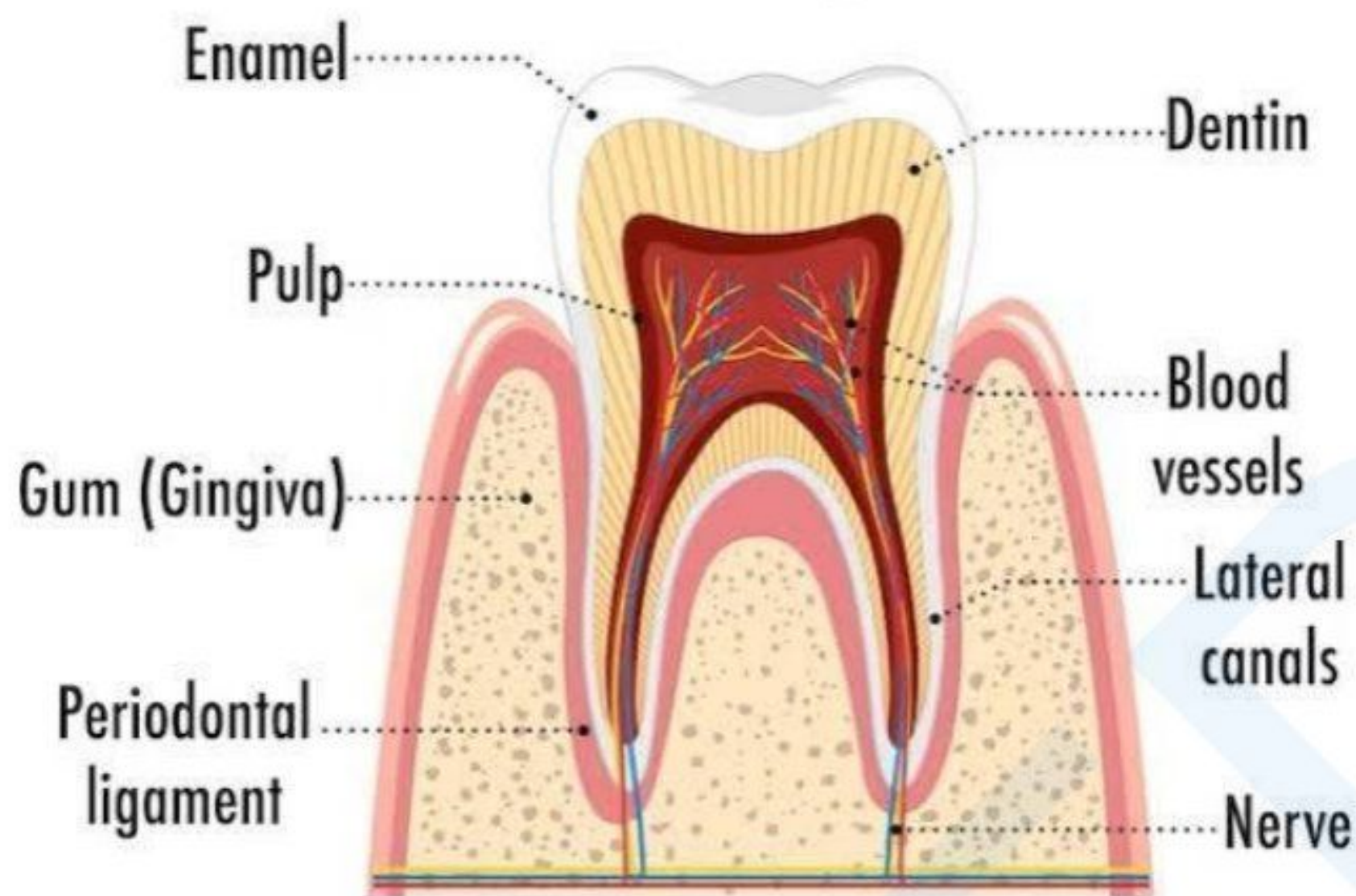


Introduction to Dental Anatomy

CROWN & ROOT

Structure of The Healthy Tooth



- **Each tooth** has a **crown & root**
- **Crown** → covered with **enamel**
- **Root** → covered with **cementum**
- **Cementoenamel junction (CEJ)** → Where **crown & root** meet (**Cervical line**)
- **Main bulk** → made of **dentin**
- **Dentin cross-section shows** → **Pulp chamber** (in crown) & **Pulp canal** (in root) → together called **pulp cavity**

TOOTH TISSUES

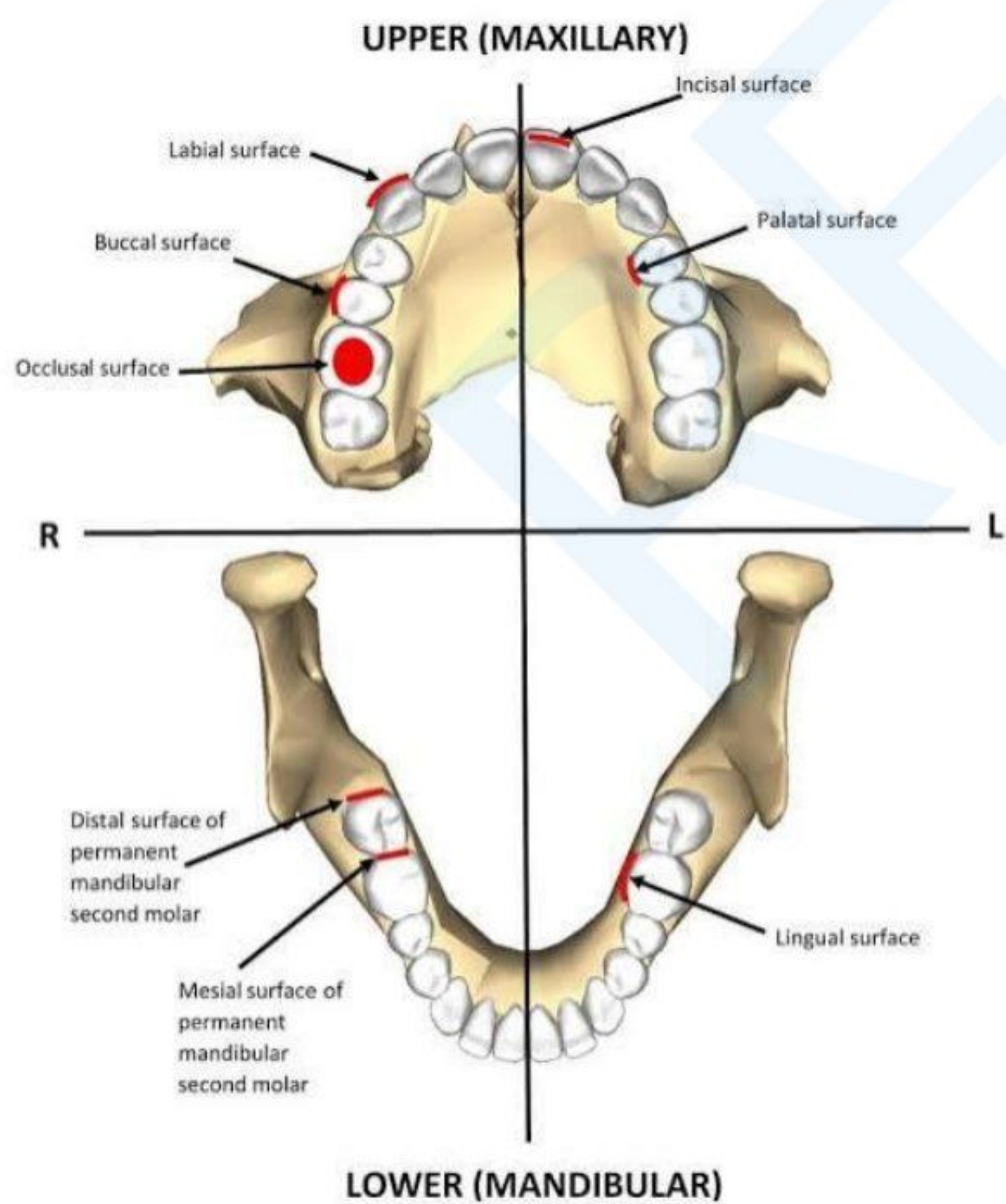
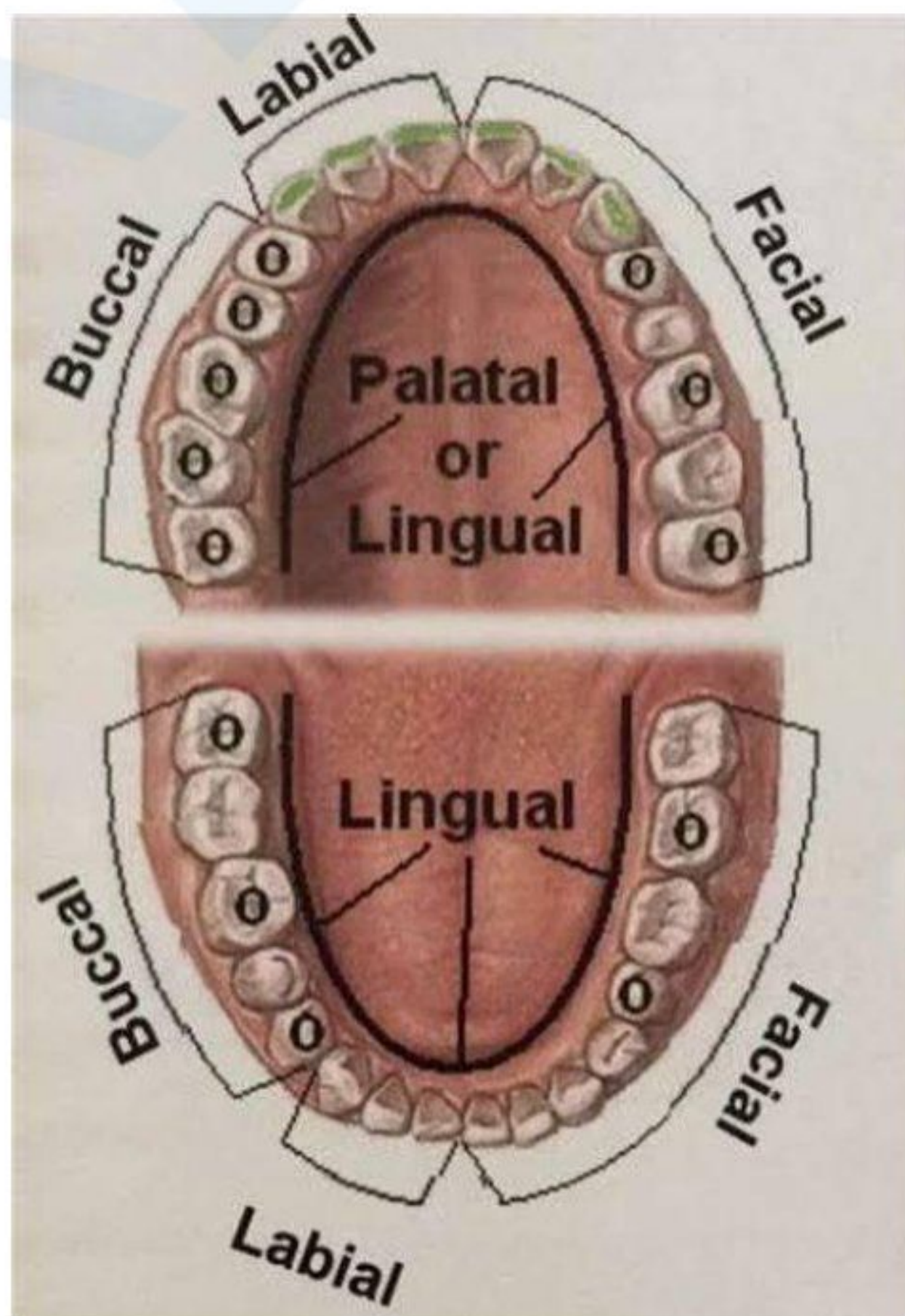
- **Four types** → **Enamel, Cementum, Dentin (Hard tissues), Pulp (Soft tissue)**
- **Pulp** → provides **blood & nerve supply**

SUPPORTING STRUCTURES

- **Alveolar process** → Part of jaw **supporting the tooth**
- **Alveolus** → **Bone of the tooth socket**
- **Gingiva (gums)** → **Covers cervical third of crown** in young adults

TOOTH SURFACES & RIDGES

- **Incisors & canines → 4 surfaces + 1 ridge**
- **Premolars & molars → 5 surfaces**



- **Surfaces based on position**

- **Labial** → Facing **lips** (incisors & canines)
- **Buccal** → Facing **cheeks** (premolars & molars)
- **Facial** → **Labial + Buccal**
- **Lingual** → Facing **tongue**
- **Occlusal** → **Chewing surface** (premolars & molars)
- **Incisal** → **Cutting edge** (incisors & canines)

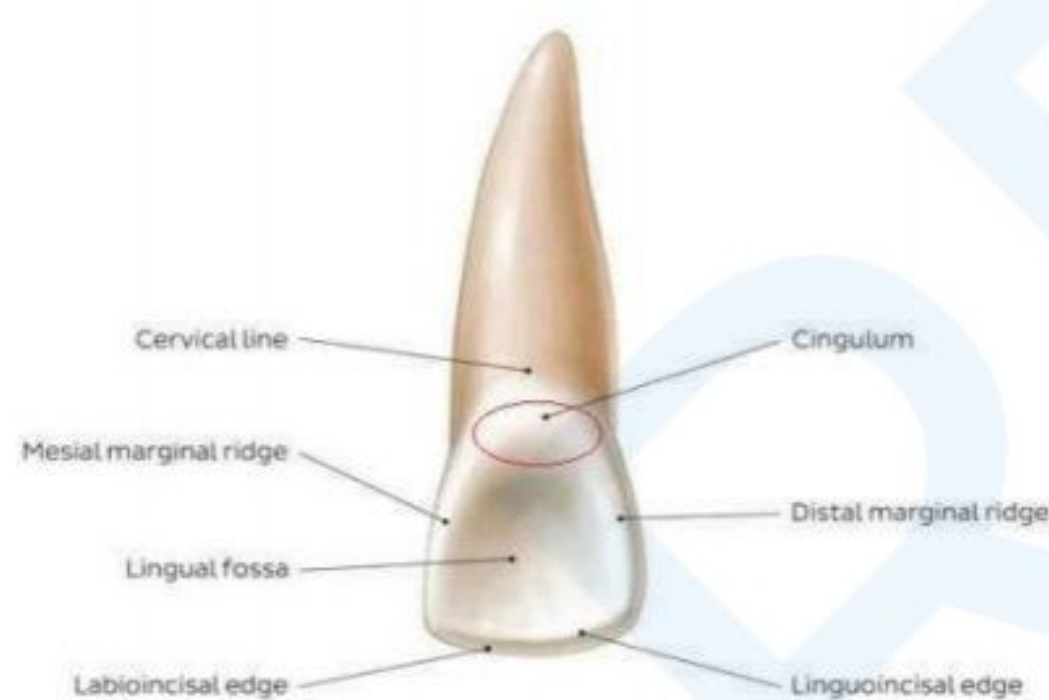
PROXIMAL SURFACES

- **Touch adjacent teeth**
 - **Mesial** → **Closer to midline**
 - **Distal** → **Away from midline**
- **Contact area** → Where **mesial/distal surface touches neighboring tooth**

TOOTH TYPES

- **Anterior teeth** → **Central & lateral incisors, Canines**
- **Posterior teeth** → **Premolars & Molars**

OTHER LANDMARKS



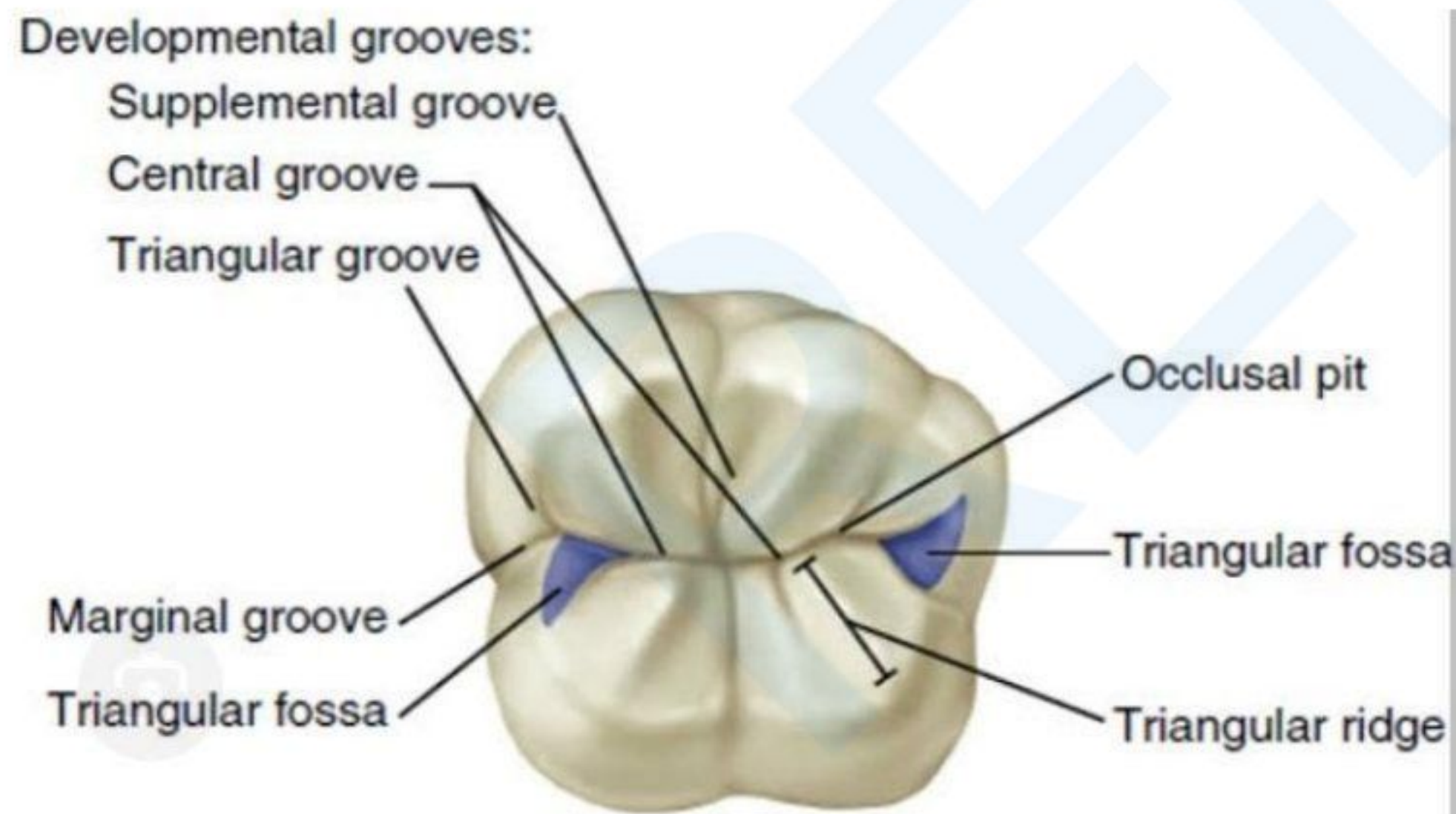
- **To study a tooth**, knowing its **important landmarks** is necessary

Elevations on the Surface of Teeth

1. **Cusp**: An **elevated mound** on the crown of the tooth, part of the **occlusal surface**.
2. **Tubercle**: A **small elevation** caused by extra enamel formation.
3. **Cingulum**: A **bulky, convex part** on the **lingual surface** of anterior teeth (incisors and canines).
4. **Ridge**: A **linear elevation** on the surface of the tooth.

- **Marginal Ridges:** Rounded enamel borders on the occlusal surface of **molars** and **premolars**.
 - **Triangular Ridge:** A **slope** descending from the cusps of **molars** and **premolars**, looking like a triangle.
 - **Transverse Ridge:** The **union** of two triangular ridges crossing the **occlusal surface**.
 - **Oblique Ridge:** A **ridge** that crosses obliquely on the occlusal surface of **maxillary molars**, formed by the **distobuccal cusp** and **mesiolingual cusp**.
5. **Fossa:** A **depression** or **concavity** on the tooth surface.
 - **Lingual Fossa:** Found on the **lingual surface** of incisors.
 - **Central Fossa:** Found on the **occlusal surface** of **molars**.
 - **Triangular Fossa:** Present **mesial** and **distal** to **marginal ridges**.
 6. **Sulcus:** A **long depression** between **ridges** and **cusps**, containing a **developmental groove**.
 7. **Lobe** → **Primary growth section** of the **crown** (**Cusps & mamelons** represent lobes)
 8. **Mamelons** → **Rounded projections** on **incisal ridges** of **newly erupted incisors**

Grooves



Depressions/Concave Features on the surface of teeth

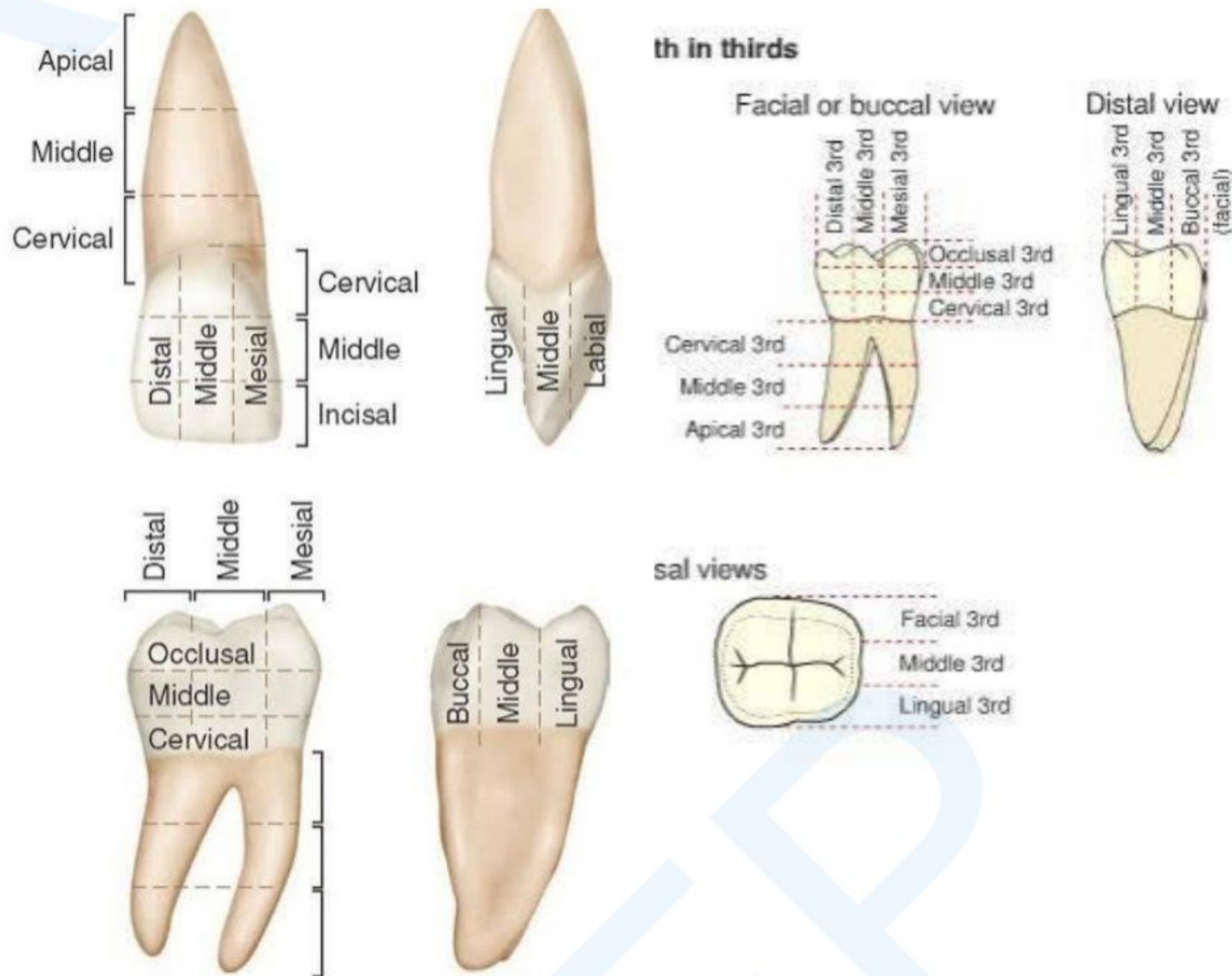
1. **Developmental Groove:** Shallow line separating the primary parts of the **crown/root**, important for the formation of **cusps** and **ridges**.

2. **Supplemental Groove:** Small, extra groove that doesn't reach **developmental grooves**.
3. **Central Groove:** Main groove running in the center of the **occlusal surface**, divides the tooth into **buccal** and **lingual parts**.
4. **Triangular Groove:** Extends from the **central groove** towards the **triangular fossa**.
5. **Marginal Groove:** Found on **marginal ridges**, allows **food passage** between teeth.
6. **Triangular Fossa:** Depressed area near triangular ridges where grooves meet.
7. **Occlusal Pit:** Small pinpoint depression where multiple grooves converge.
8. **Pits:** Pinpoint depressions at the junction of **developmental grooves** (e.g., **central pit**).

TOOTH ROOTS

- **Single-rooted** → All anterior teeth, Mandibular 1st & 2nd premolars, Maxillary 2nd premolar
- **Double-rooted** → Maxillary 1st premolar (Buccal & Lingual root), Mandibular molars (Mesial & Distal root)
- **Triple-rooted** → Maxillary molars (Mesiobuccal, Distobuccal & Lingual root)

DIVISION INTO THIRD



- **Crown** and **Root** are divided into **thirds** for description
- **Crown** is divided into: **Incisal/Occlusal third, Middle third, Cervical third**
- **Root** is divided into: **Cervical third, Middle third, Apical third**
- **Crown can be divided in 3 ways:**
 - **Inciso/Occluso-Cervically**
 - **Mesiodistally**
 - **Labio/Bucco-Lingually**

LINE ANGLES

- **Formed by the junction of two surfaces**
- Named by combining the two surfaces that meet

LINE ANGLES OF ANTERIOR TEETH

- **Mesiolabial**
- **Distolabial**

- Mesiolingual
- Distolingual
- Labioincisal
- Linguoincisal

LINE ANGLES OF POSTERIOR TEETH

- Mesio Buccal
- Distobuccal
- Mesiolingual
- Distolingual
- Mesio-occlusal
- Disto-occlusal
- Bucco-occlusal
- Linguo-occlusal

POINT ANGLES

- Formed by the junction of three surfaces

POINT ANGLES OF ANTERIOR TEETH

- Mesiolabioincisal
- Distolabioincisal
- Mesiolinguoincisal
- Distolinguoincisal

POINT ANGLES OF POSTERIOR TEETH

- Mesio Bucco-occlusal
- Distobucco-occlusal
- Mesiolinguo-occlusal
- Distolinguo-occlusal

TOOTH DRAWING AND CARVING

- Drawing and carving of teeth is done **along with lectures** on dental anatomy
- **G. V. Black's table of average measurements** is used as a base but **modified for better articulation**

- **Teeth are carved in natural size** with minimal fractions (only 0.5 mm & 0.3 mm used)

PURPOSE OF CARVING AND MEASUREMENTS

- **Helps in comparing tooth dimensions**
- **Maxillary central incisor is 8.5 mm mesiodistally**, canines are **1 mm narrower**
- **Tooth outlines are recorded on squared-millimeter paper** for accuracy

TOOTH ASPECTS FOR COMPARISON

- **Mesial**
- **Distal**
- **Labial/Buccal**
- **Lingual**
- **Incisal/Occlusal**

SKILLS TO DEVELOP

- **Memorize measurement table** for quick mental comparisons
- **Draw accurate outlines** of any tooth aspect
- **Carve designs with precision** matching line drawings

MEASURING TEETH

- **Use of Boley gauge** is essential
- **Each tooth has 8 calibrations** to measure key landmarks
- **Different methods for anterior & posterior teeth** are used

1. Teeth Categories

- Maxillary Teeth
- Mandibular Teeth

2. Measurement Parameters

- **Length of Crown**
- **Length of Root**
- **Mesiodistal Diameter of Crown**
- **Mesiodistal Diameter at Cervix**
- **Labiolingual/Buccolingual Diameter of Crown**
- **Labiolingual/Buccolingual Diameter at Cervix**

- **Curvature of Cervical Line (Mesial & Distal)**

3. Individual Teeth Specifications (for both maxillary & mandibular)

- Central Incisor
- Lateral Incisor
- Canine
- First Premolar
- Second Premolar
- First Molar
- Second Molar

- Third Molar

TABLE 1-1 Measurements of the Teeth: Specifications for Drawing and Carving Teeth of Average Size*

	LENGTH OF CROWN	LENGTH OF ROOT	MESIODISTAL DIAMETER OF CROWN [†]	MESIODISTAL DIAMETER OF CROWN AT CERVIX	LABIO- OR BUCCOLINGUAL DIAMETER OF CROWN	LABIO- OR BUCCOLINGUAL DIAMETER OF CROWN AT CERVIX	CURVATURE OF CERVICAL LINE— MESIAL	CURVATURE OF CERVICAL LINE—DISTAL
Maxillary Teeth								
Central incisor	10.5	13.0	8.5	7.0	7.0	6.0	3.5	2.5
Lateral incisor	9.0	13.0	6.5	5.0	6.0	5.0	3.0	2.0
Canine	10.0	17.0	7.5	5.5	8.0	7.0	2.5	1.5
First premolar	8.5	14.0	7.0	5.0	9.0	8.0	1.0	0.0
Second premolar	8.5	14.0	7.0	5.0	9.0	8.0	1.0	0.0
First molar	7.5	B L 12 13	10.0	8.0	11.0	10.0	1.0	0.0
Second molar	7.0	B L 11 12	9.0	7.0	11.0	10.0	1.0	0.0
Third molar	6.5	11.0	8.5	6.5	10.0	9.5	1.0	0.0
Mandibular Teeth								
Central incisor	9.0 [‡]	12.5	5.0	3.5	6.0	5.3	3.0	2.0
Lateral incisor	9.5 [‡]	14.0	5.5	4.0	6.5	5.8	3.0	2.0
Canine	11.0	16.0	7.0	5.5	7.5	7.0	2.5	1.0
First premolar	8.5	14.0	7.0	5.0	7.5	6.5	1.0	0.0
Second premolar	8.0	14.5	7.0	5.0	8.0	7.0	1.0	0.0
First molar	7.5	14.0	11.0	9.0	10.5	9.0	1.0	0.0
Second molar	7.0	13.0	10.5	8.0	10.0	9.0	1.0	0.0
Third molar	7.0	11.0	10.0	7.5	9.5	9.0	1.0	0.0

B, Buccal; L, lingual.

*In millimeters. This table has been "proved" by carvings shown in Figures 1-16 and 1-17.

†The sum of the mesiodistal diameters, both right and left, which gives the arch length, is maxillary, 128 mm; mandibular, 126 mm.

‡Lingual measurement is approximately 0.5 mm longer.

TABLE 1-1 Measurements of the Teeth: Specifications for Drawing and Carving Teeth of Average Size—cont'd

MEASUREMENTS OF THE TEETH: AN EXAMPLE*

	LENGTH OF CROWN	LENGTH OF ROOT	MESIODISTAL DIAMETER OF CROWN [†]	MESIODISTAL DIAMETER OF CROWN AT CERVIX	LABIO- OR BUCCOLINGUAL DIAMETER OF CROWN	LABIO- OR BUCCOLINGUAL DIAMETER OF CROWN AT CERVIX	CURVATURE OF CERVICAL LINE—MESIAL	CURVATURE OF CERVICAL LINE—DISTAL
Maxillary Teeth								
Central incisor	10.5	13.0	8.5	7.0	7.0	6.0	3.5	2.5

*In millimeters.

†The sum of the mesiodistal diameters, both right and left, which gives the arch length, is maxillary, 128 mm; mandibular, 126 mm.